

13.8 A 1a-nm 1.05V 10.5Gb/s/pin 16Gb LPDDR5 Turbo DRAM with WCK Correction Strategy, a Voltage-Offset-Calibrated Receiver and Parasitic Capacitance Reduction

Publisher: IEEE [Cite This](#)

<< Results | < Previous | Next >

[Yangho Seo](#) ; [Jihee Choi](#) ; [Sunki Cho](#) ; [Hyunwook Han](#) ; [Wonjong Kim](#) ; [Gyeongha Ryu](#) 

9

Cites in Papers

2847

Full Text Views

Abstract
Authors
Figures
References
Citations
Keywords
Metrics
More Like This



Down
PDF

Abstract:
The LPDDR product family originally sought to minimize power consumption. As the LPDDR5X is released with a 33% increase in maximum operating speed, low-power and high-sp... [Show More](#)

▼ Metadata

Recommended for You (Beta)

5-nm High-Efficiency and High-Density Digital SRAM In-Memory-Computing Macros...

Dyamond: Compact and Efficient 1T1C DRAM IMC Accelerator With Bit Column Addition for...

Reconfigurable Precision INT4-8/FP8 Digital Compute-in-Memory...

[Learn More](#)

 Contents

The LPDDR product family originally sought to minimize power consumption. As the LPDDR5X is released with a 33% increase in maximum operating speed, low-power and high-speed operations have become essential [1]. Moreover, expanding virtual space (AR, XR) and the artificial intelligence industry have accelerated demand for high-speed and low-power mobile DRAM products. Therefore, guaranteeing higher IO speed has become a significant concern in designing DRAM. This paper proposes a WCK correction strategy, a voltage-offset-calibrated receiver, and IO for DRAM test to achieve high-speed operation. The WCK correction strategy improves the 3-sigma 4-phase skew distribution by 65%. The offset-calibrated receiver with a 1-tap decision-feedback equalizer (DFE) enhances the voltage offset by 59%. IO for DRAM test reduces the parasitic capacitance of the DQ pad by up to 39%. Using these techniques for high-speed operation, the LPDDR5X achieves operating speed up to 10.5Gb/s at $V_{DD2H} = 1.05V$ and 10.0Gb/s at $V_{DD2H} = 0.95V$.

Figure 13.8.1 shows the overall architecture of the LPDDR5X with the proposed features. For higher speed operation, WCK must have minimum duty error and 4-phase skew, which both directly affect read and write timing margins; hence, a strategy to get a near ideal 4-phase WCK is necessary. A duty-cycle corrector (DCC) in the WCK buffer, duty cycle monitor (DCM) and duty cycle adjuster (DCA), and 4-phase delay trimming circuit are used to achieve 4-phase alignment. The trimmed 4-phase WCK is provided to the DQ RX, which uses voltage-offset calibration and a 1-tap DFE. The offset calibration uses ± 3 steps and operates each DQ sequentially. Once this calibration is complete, all receivers have reduced errors which is caused by process



variation. When it comes to mass production, a DRAM test must be performed before packaging. To test a DRAM without impacting the DQ pad, the IO pad for the DRAM test must be isolated from the DQ pad.

Three significant factors cause WCK SI degradation: external duty error, internal duty error, and 4-phase skew. The external duty error is duty error caused outside of the DRAM, while the internal duty error is generated inside the DRAM. Because duty error and skew propagate to subsequent stages and distort the signal, the duty error should be corrected at each stage. When the controller and channel cause an external duty error, the self-DCC scheme in the WCK buffer mitigates this error [2]. This scheme corrects the output duty error within $\pm 4\%$ when the input WCK duty is 43 - 57% at 5.25GHz. The rest of the external and internal duty errors are calibrated by DCM/DCA training. To make this training successful a high-resolution DCM is needed. Figure 13.8.2 shows the DCM with adaptive-enable logic to improve its resolution. Once either output nodes of the DCM (OUT or OUTB) become lower than the logic threshold, the output of this logic (LAT_EN) goes high. The logic guarantees the optimum operation time to achieve maximum ΔV , so it helps to maintain the DCM high-resolution regardless of PVT skew.

After minimizing the duty error, the 4-phase skew remains in the 4-phase WCK path, which is caused by process skew. A trimming sequence, after fabrication, helps to improve the distribution of the skew. Unlike conventional ring oscillator delay (ROD), a single-edge ROD needs only the rising edge to run. This ROD generates a falling edge after the rising edge automatically, so only the rising edge propagation delay difference can be calculated. Because the definition of the 4-phase skew has nothing to do with the falling edge, this ROD helps to improve the measurement accuracy by up to 58%. A phase-averaging scheme mixes each phase with its 180° different phase to reduce the rest of the skew, as shown in Fig. 13.8.2.

Figure 13.8.3 shows the 1-tap DFE and the capacitive load-controlled offset calibration scheme implemented in a double-tail latch to reduce inter-symbol interference (ISI) and mitigate offset. A double-tail latched comparator has no static power dissipation [3]. In addition, a direct DFE will be implemented with low power consumption and a simple structure compared to a loop-unrolled DFE. The DFE changes the tap coefficient by adjusting the transconductance (gm) of the two input transistors; thus, the output voltage (OIP and OIN) is improved based on the previous decision. Two sets of capacitors are connected to the OIP and OIN nodes to perform the RX voltage-offset calibration. When the offset calibration signal (OCC_EN) is enabled, an internal counter selects all BLs and DQs sequentially, and the offset is compensated by changing OSDN[0:1] and OSUP[0:1].

There are two purposes of the DQ pad: one is packaging, and the other is testing before packaging. If a DRAM test and packaging is done on the same pad, there is a risk that the packaging is not successful due to defects made during testing. Therefore, an additional pad for testing is needed, attached to the pad for package. However, this test pad causes speed degradation because it increases capacitance of the DQ pins. The capacitance of a pad is the most significant among the components before the proposed IO path is implemented. Separating test pad from the DQ pad for packaging achieves reduced capacitance on a pad, and it needs a simple transmitter and receiver circuit. Figure 13.8.4 shows how the new data path connects to the pad for testing instead of the DQ pad. The TX circuit for the test operates as an output driver and on-die termination (ODT), and it transmits a signal branched from the normal read path to the test pad. An RX for the test receives the signal from the test pad, and it is finally delivered to the RX in the DQ pad through the TX connected to DQ pad.

Figure 13.8.5 shows the results of implemented schemes. The offset is reduced by 59% compared to that without the offset calibration scheme. Moreover, separating pads achieves a 39% capacitance reduction on the DQ pad: improving the operating speed. Figure 13.8.6 shows the ATE measured results of t_{CK} Shmoo with various voltage and clock periods and the eye Shmoo with the write and read operations. The implemented design achieves maximum speed as 10.5Gb/s using a 1.05V V_{DD2H} and 10Gb/s using 0.95V. The write and read valid windows are 0.79 and 0.76UI: 1UI is 95.2ps. Figure 13.8.7 shows the micrograph of fabricated DRAM of 1a-nm process occupying 51.3mm²/channel (16Gb).



Figure 13.8.1:
Interconnection diagram between SoC and DRAM.



Figure 13.8.2:
WCK block diagram with DCC, adaptive DCM, phase-averaging, and single-edge ROD.



Figure 13.8.3:
Receiver with 1-tap DFE and offset calibration sequence.



Figure 13.8.4:
Implemented IO path for DRAM test. This circuit is needed to transfer the normal WR and RD data for the test.



Figure 13.8.5:
(a) Offset calibration result by Monte-Carlo simulation, (b) reduced C_{IO} , and (c) WCK skew error.



Figure 13.8.6:
Measured t_{CK} and WR/RD Shmoo plots.



Figure 13.8.7:
Chip micrograph.

